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2.5

MH1200 Linear Algebra I.

The 2nd Quiz, 2023.

- Please write down your **name** and **matriculation number** at the top. The duration is **22 minutes**. There are **two** questions. A total of **5** marks.
- You may write on every part of this document and also use extra paper if you need. To get full marks please write down all of your working.

Problem 1: (3 marks.)

The following is the augmented matrix of a system of linear equations in three variables x, y and z depending on a parameter a. Determine for which values of the parameter a the system has 0 solutions, 1 solution, and infinitely many solutions.

$$\left[\begin{array}{ccc|c} a+1 & 0 & 1 & 1 \\ 0 & a-1 & 1 & a \\ 0 & a-1 & a^2 & a^3+a-1 \end{array}\right]$$

$a^3-a^2 \cancel{+a-1} -a+1$

$\therefore a^3-a^2$

$$\left[\begin{array}{ccc|c} a+1 & 0 & 1 & 1 \\ 0 & a-1 & 1 & a \\ 0 & a-1 & a^2 & a^3+a-1 \end{array}\right] \xrightarrow{C_1 \leftrightarrow C_3} \left[\begin{array}{ccc|c} 1 & 0 & a+1 & 1 \\ 1 & a-1 & 0 & a \\ a^2 & a-1 & 0 & a^3+a-1 \end{array}\right] \xrightarrow[R_3 \to R_3 - a^2R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 0 & a+1 & 1 \\ 0 & a-1 & -a-1 & a-1 \\ 0 & a-1 & -a^2(a+1) & a^3-a^2+a-1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & a+1 & 1 \\ 0 & a-1 & -a-1 & a-1 \\ 0 & a-1 & -a^2(a+1) & a^3-a^2+a-1 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & a+1 & 1 \\ 0 & a-1 & -a-1 & a-1 \\ 0 & 0 & -a^3-a^2+a+1 & a^3-a^2 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} a+1 & 0 & 1 & 1 \\ 0 & a-1 & 1 & a \\ 0 & a-1 & a^2 & a^3+a^2-1 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} a+1 & 0 & 1 & 1 \\ 0 & a-1 & 1 & a \\ 0 & 0 & a^2-1 & a^3-1 \end{array}\right]$$

When $a=1$,

$$\left[\begin{array}{ccc|c} \neq 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore$ There are infinitely many solutions.

When $a=-1$,

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 0 & \neq 0 & 1 & -1 \\ 0 & 0 & 0 & \neq 0 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_1} \left[\begin{array}{ccc|c} 0 & -2 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -2 \end{array}\right]$$

There are no solutions since it is an inconsistent systems.

When $a=0$,

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{array}\right]$$

There is a unique solution.

When $a \neq 0, 1, -1$

$$\left[\begin{array}{ccc|c} \neq 0 & 0 & 1 & 1 \\ 0 & \neq 0 & 1 & \neq 0 \\ 0 & 0 & \neq 0 & \neq 0 \end{array}\right]$$

There is a unique solution.

~~For the system to have~~

When $a=1$,

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

There are infinitely many solutions when $a=1$, as the ~~2n~~ second column has no leading entry, so free parameters are introduced.

When $a=-1$

$$\left[\begin{array}{ccc|c} 1 & 0 & \cancel{4}0 & 1 \\ 0 & -2 & 0 & \cancel{a-1}\ -2 \\ 0 & 0 & 0 & -2 \end{array}\right]$$

~~This is inconsistent, as the ti last row~~

The last row is inconsistent. Hence, there are no solutions when $a=-1$.

When $a=0$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -1 & -1 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

This is in row echelon form and every column has a leading entry. Hence the system has 1 solution when $a=0$.

-1

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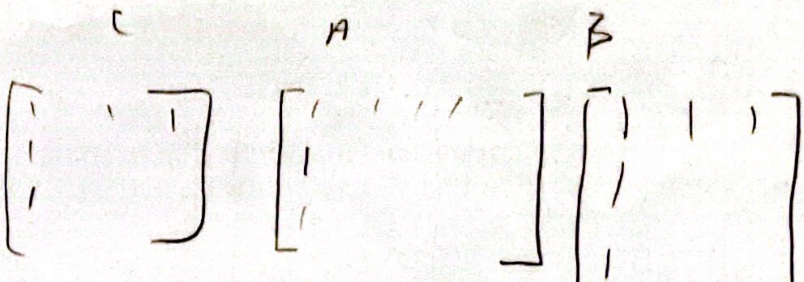

Problem 2: (2 marks.)

Let $\mathbf{A} = [a_{ij}]_{3\times4}$, $\mathbf{B} = [b_{ij}]_{4\times3}$ and $\mathbf{C} = [c_{ij}]_{3\times3}$ be three matrices.

(i) Give an expression for the entry $(3,1)$ of the matrix $\mathbf{CAB}$ in terms of the entries of these matrices.

(ii) Identify a matrix expression using $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$, and an entry of the resulting matrix, which is given by the following formula:

$$\sum_{k=1}^{3} a_{k4}c_{k2} - \sum_{l=1}^{3} b_{4l}c_{2l}.$$

i. (3,1) entry of CAB $= \sum_{k=1}^{4}\left(\sum_{z=1}^{3} c_{3z}\, a_{z1}\right) b_{1k}$ $-\delta$

ii.

i. The (3,1)-entry of CAB
$= \sum_{l=1}^{3}\sum_{k=1}^{4} c_{3l}(a_{lk}b_{k1})$

ii. $\sum_{k=1}^{3} a_{k4}c_{k2} - \sum_{l=1}^{3} b_{4l}c_{2l}$
This is the (4,2) entry of A^TC-BC^T